Identification of Forest Ground and Canopy Peaks From 3-D SAR Tomographic Profile Using Deep Learning

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***Abstract*—Tomographic synthetic aperture radar (TomoSAR) at low frequency, i.e., P-/L-band, has become a promising tool for forest structure study. Forest canopy height and underlying topography are two of the most important parameters one can estimate using the TomoSAR technique. One simple way to estimate these two parameters is to detect the peaks of the tomographic profile, which, however, can lead to large biases due to complicated forest structure, sidelobes, or insufficient TomoSAR resolution. Polarimetric TomoSAR (Pol-TomoSAR) provides a solution to this by exploring the polarimetric diversity to separate the ground and canopy components and then conducting an independent TomoSAR analysis. However, the Pol-TomoSAR technique suffers from a low ground-to-volume ratio (GVR), which often leads to unsuccessful ground and canopy separation. To mitigate this problem, in this article, we provide a deep-learning solution to ground and canopy height estimation from a 3-D tomographic profile through the identification of the patterns of ground and canopy peaks. A 3-D U-net model is introduced in our solution to grasp as many 3-D characteristics of the tomographic profile as possible. Moreover, our model can be well trained using only a synthetic TomoSAR dataset, making it easy to implement when we don't have enough real data with LiDAR References. The proposed method is validated on P-band real TomoSAR dataset from multiple test sites in AfriSAR campaign, showing that it can achieve more accurate ground and canopy height estimation than the state-of-the-art Pol-TomoSAR techniques. The maximum root-mean-square errors (RMSE) improvement reaches as high as 66.4% and 63.2% for ground and canopy top height, respectively.**

***Index Terms*—3-D U-net, canopy height, deep learning, forest, ground, tomographic synthetic aperture radar (TomoSAR).**

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I. Introduction

TOMOGRAPHIC synthetic aperture radar (TomoSAR) [1], [2] at low frequency, i.e., P-/L-band, has now become a promising tool for forest structure study because it can penetrate into the forest and even "see" the underlying ground. Unlike LiDAR's spatially discrete sampling and dependence on weather conditions, TomoSAR can provide continuous wall-to-wall measurement with comparable accuracy in all weather conditions.

Digital terrain model (DTM) and canopy height model (CHM) are two of the most direct and important parameters one can estimate from TomoSAR measurements. One way to estimate DTM and CHM from the tomographic profile is to simply detect the two largest peaks [3], [4], which, however, is easily to be affected by sidelobes and insufficient TomoSAR vertical resolution, or complicated forest structures with multiple peaks between the ground and the canopy layers.

Polarimetric TomoSAR (Pol-TomoSAR) provides a solution to this by exploring the polarimetric observation diversity to separate the ground and volumetric components and conduct TomoSAR DTM and CHM estimation afterward. An algebraic synthesis method was first proposed in [5] and [6] to separate the ground and volumetric components using the sum of Kronecker product (SKP) decomposition, which was later applied for DTM estimation on P-band tropical forest full-pol tomographic dataset [7]. A model-based SKP decomposition method was also introduced in [8] to perform a three-component separation, including double bounce scattering, odd bounce scattering, and volumetric scattering. However, these methods often fail in ground separation in low ground-to-volume ratio (GVR) areas where the ground scattering power is low [9]. This problem is partially solved by [9], which brings a maximum likelihood separation of the ground and volumetric component by introducing parameterized ground and volumetric polarimetry models. Nevertheless, unsuccessful separations still happen at rather low GVR, which leads to biased DTM and CHM estimation.

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In essence, the estimation of DTM and CHM from TomoSAR measurement lies in the identification of ground and canopy peak patterns. However, the current methods mentioned above cannot provide accurate and robust identification due to low GVR values. In terms of pattern recognition of the ground and canopy peak, deep learning is a quite suitable choice by virtue of its excellent performance in feature extraction, representation, and pattern learning, which is now widely spread in nearly all walks of scientific research [10]. It has been introduced into TomoSAR research in several studies either to enhance the tomographic profile [11], [12] or learn DTM/CHM from full-pol TomoSAR image stack [13], [14]. In [13] and [14], the model is trained on a real multi-Pol-TomoSAR dataset with LiDAR measurements as label data. Applying such a method to data acquired with observation geometries different from those of the training scenario may be critical. To overcome this generalization problem, in this article, we introduce a deep-learning model that learns DTM and CHM directly from 3-D tomographic profiles by identifying ground and canopy peak locations and can be well-trained on just single-pol simulated dataset. Once the model training is finished, it can be applied to different test sites of different observation geometries without making changes. Aimed at grasping as many 3-D spatial features of the 3-D tomographic profile as possible and thereby achieving more accurate ground and canopy peak identification, a 3-D U-net structure [15] is selected, given its excellent capability in multiscale 3-D feature extraction. Two essential adaptations are made to ensure that the 3-D U-net model works well in the context of ground and canopy peak identification from a 3-D tomographic profile. One of them is the label-filling strategy, which fills the internal space between the ground and canopy peak locations with ones. This strategy can largely prevent false identification between the ground and canopy peak locations. Another one is the maximum absolute error (MAE) loss function, which ensures a sharp prediction map and thereby a less biased ground identification. To train the 3-D U-net model, TomoSAR datasets are first simulated based on prior ground and canopy peak locations using the 2-D integration model [16], which are then transformed into the spectral domain using Capon beamforming [17] to generate 3-D tomographic profiles. Since the purpose of the 3-D U-net model is to identify the ground and canopy peaks from 3-D tomographic profiles, these 3-D profiles generated by the simulated dataset are then fed into the 3-D U-net model as training data, and the true ground and canopy peak locations serve as label data with the help of a label-filling strategy. Compared with the Pol-TomoSAR techniques and some of the deep learning ones, the proposed method requires only single-pol data. Besides, our deep-learning network can be well trained using only a simulated dataset, which makes it simpler to implement if we do not have enough real TomoSAR data with LiDAR references.

The remainder of this article is structured as follows. Section II introduces the 3-D U-net model for ground and canopy peak identification, the TomoSAR data simulation through the 2-D integration model, and the label-filling strategy. Model training details and results, and numerical experiments on multiple real airborne TomoSAR datasets are conducted in Section III, and conclusions are drawn in Section IV.

II. Methodology

Fig. 1 demonstrates the whole workflow of the proposed deep-learning method, including the dataset simulation and model training. To train our model, we proposed to feed it with the 3-D tomograms generated by a simulated TomoSAR dataset as training data. For the TomoSAR dataset simulation, based on the prior canopy peak height Z_v and underlying topography Z_g, the 3-D tomographic backscattering profile along the height direction is generated using the weighted sum of two Gaussian functions centered on Z_v and Z_g. Then the 3-D profile serves as the input for TomoSAR dataset stack simulation based on the 2-D integration model [16]. After that, the simulated TomoSAR data stack is transformed into the spectral domain using Capon beamforming to generate 3-D tomograms, which, together with Z_g and Z_v, are then served as the input data and labels to train the 3-D U-net model. The 3-D U-net model and parameter settings, and the simulation of the TomoSAR dataset will be elaborated in detail in the following.

A. 3-D U-Net Model for Ground and Canopy Peak Identification

Three-dimensional U-net model was first introduced in [15] for biomedical images segmentation. The encoder–decoder U-net structure makes it sensitive to both local and global features of an image [18]. It has become one of the baseline models in the context of image segmentation. Unlike the 2-D U-net model [18], the 3-D U-net model uses 3-D convolution, allowing it to grasp more 3-D features of a 3-D biomedical image. Therefore, for any image objects with 3-D patterns, the 3-D U-net is naturally the better choice. TomoSAR at low frequency takes 3-D measurements of natural semi-transparent targets like forests, icebergs, etc., and generates 3-D tomograms that reflect the internal structure of these targets. The 3-D U-net model is therefore chosen for our deep-learning method for ground and canopy peak identification.

Fig. 2 shows the network structure of the 3-D U-net model used in our work. To keep the model lightweight, only four downsampling encoder layers are constructed. Correspondingly, four upsampling decoder layers are constructed, and four skip connections are constructed to concatenate both local and global features from different layers. Each downsampling layer contains one 3-D convolution block followed by one maximum pooling block, and each 3-D convolution block consists of two concatenated 3-D convolution operations with batch normalization (BN) and rectified linear unit (ReLU) activation function. Each upsampling layer contains one 3-D deconvolution (or upsampling) block followed by one 3-D convolution block. The input and output channels are set to one for our 3-D tomogram input and predicted map. Another two critical points are the label-filling strategy and the MAE loss function, which will be elaborated in detail in the following.

In this article, a label-filling strategy is applied to the label data. In this strategy, both the ground and canopy peak

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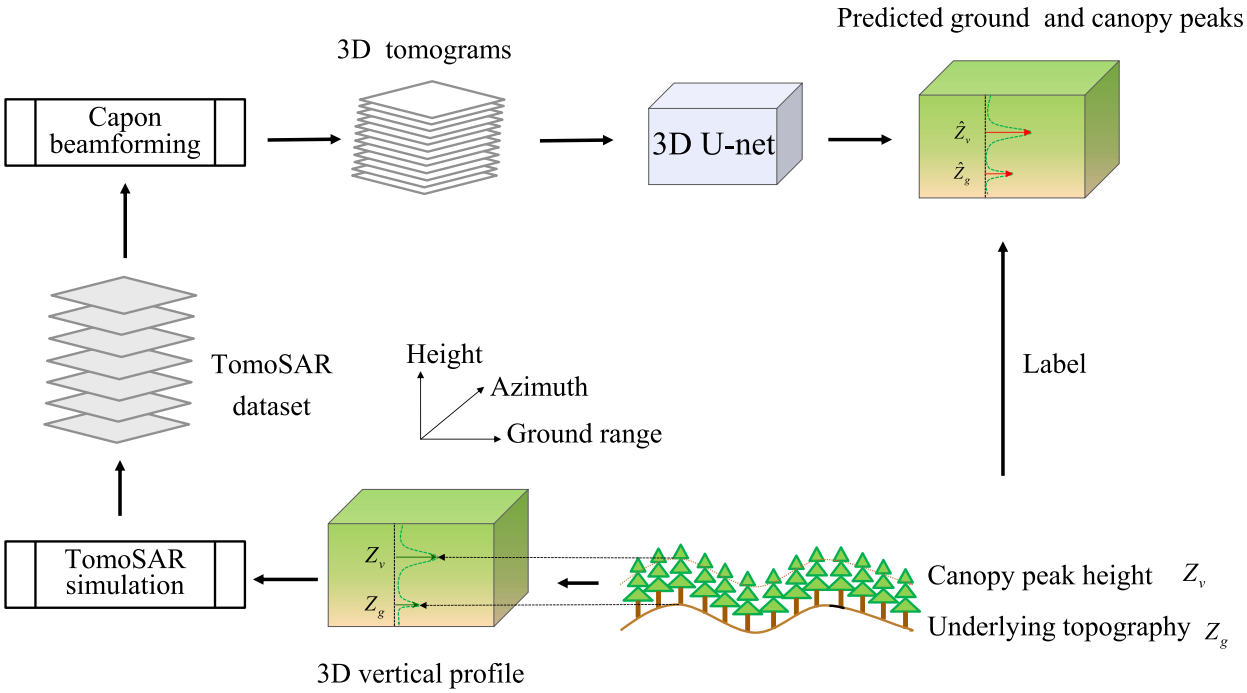


Fig. 1. Flowchart of the proposed methodology.

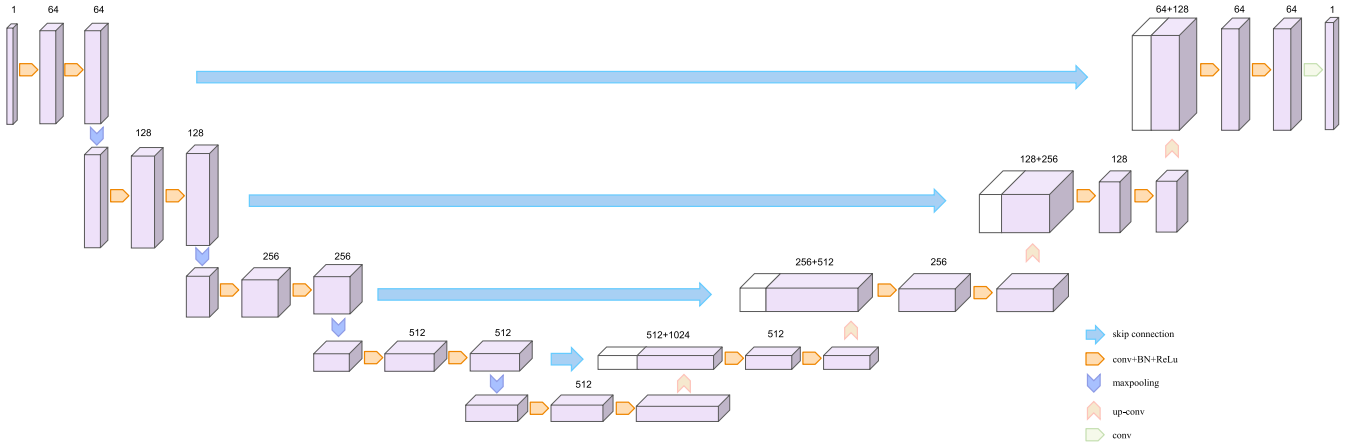


Fig. 2. Structure of the 3-D U-net model for ground and canopy peaks identification.

locations and the internal space between them are filled with ones, and the other locations are set to zeros. If we are to learn the ground and canopy peak locations from the 3-D tomograms, the simplest way is to label the true ground and canopy peak locations as ones and the others as zeros, and then train the model to learn these two peak locations. It may work if there are only two peaks in the 3-D tomogram corresponding to the ground and canopy peaks. Unfortunately, there are usually multiple peaks between or outside the ground peak and canopy peak caused by complicated forest internal structure or sidelobes. These nonground/canopy peaks will cause strongly biased ground and canopy peak estimation. As mentioned above, the U-net structure is capable of learning both local and global features simultaneously through its multiple encoder layers and skip connection structure. The location of ground and canopy peaks is just a local feature. Therefore, rather than just labeling the ground and canopy peak locations as one to learn only the local features, we propose labeling the internal locations between the ground and canopy peaks with ones as well. They can be treated as a global feature that indicates the valid region of forest backscattering along the height direction. This operation has proved to be vital for the 3-D U-net model to perform accurate ground and canopy peak prediction (see in Section III-B).

Another point to be emphasized is the loss function. In the original 3-D U-net model, the cross-entropy loss function is used because of its excellent performance in image segmentation. In this article, however, we deem the learning process as a regression mission, which means the cross-entropy loss function is no longer suitable. For a regression

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TABLE I
BASELINE CONFIGURATION AND RADAR SYSTEM PARAMETERS IN SIMULATION

Paramter	Baseline spacing (m)	Number of images	Flight height (m)	Elevation spacing (m)	Ground range spacing (m)	Azimuth spacing (m)	Center frequency (GHz)	Incident angle (degree)
Value	15.0	7	4000	3.0	2.0	1.2	397.5	45.0

mission, normally, the mean square error (mse) and MAE loss functions are commonly used. In this article, we select the MAE loss function because it can give a much sharper prediction map than mse and thereby achieve more accurate ground identification. The MAE loss function can be expressed as

$$\text{MAE_loss} = \frac{1}{\text{MNH}} \sum_{k}^{k=H} \sum_{j}^{j=N} \sum_{i}^{i=M} \left| \hat{T}_{i,j,k} - T_{i,j,k} \right| \tag{1}$$

where $\hat{T}_{i,j,k}$ is the predicted 3-D map by the 3-D U-net model and $T_{i,j,k}$ is the 3-D label. $H \times N \times M$ is the 3-D size of a 3-D tomogram sample.

B. TomoSAR Dataset Simulation

Normally, to train the designed 3-D U-net model, a large amount of TomoSAR dataset with high-quality LiDAR measurements is needed, which, except for a few well-known campaigns like TropiSAR and AfriSAR, is not easily accessible. Therefore, in this article, we train our model with a simulated dataset. As shown in Fig. 1, the 3-D vertical backscattering profile is first synthesized using the prior canopy peak height and underlying topography, which then serves as the input for TomoSAR dataset simulation through the 2-D integration model [16].

The simulation starts with prior canopy peak height Z_v and underlying topography Z_g, which are adopted from the LiDAR-measured CHM and DTM of the Lope test site in the AfriSAR campaign. Similar to [19], under the assumption of two-layer forest structure, the 3-D backscattering profile along the height direction is first synthesized by the weighted sum of two gaussian functions centered on Z_v and Z_g, which can be expressed as

$$s(z) = \frac{1}{\sqrt{2\pi}\sigma_v} e^{\frac{(z-z_v)^2}{2\sigma_v^2}} + \omega \cdot \frac{1}{\sqrt{2\pi}\sigma_g} e^{\frac{(z-z_g)^2}{2\sigma_g^2}} \tag{2}$$

where σ_g and σ_v are parameters that control the shape of the two Gaussian functions. σ_g is set to be smaller than σ_v as the ground scattering power distribution is supposed to be more concentrated than the canopy volumetric scattering power distribution. ω is a weighting factor that can reflect the GVR. A larger ω means more ground power, thus a higher GVR and vice versa. To imitate a complicated forest structure scene, the weighting factor ω is set to be a uniformly distributed random value that varies between 0.3 and 2.0.

Although the two-layer model assumption is applied in TomoSAR data simulation, it is not the precondition of our proposed deep-learning model. Our proposed deep-learning model is still applicable for DTM and CHM estimation in tropical forests. Indeed, there can be multiple peaks in the tomographic profile of tropical forests due to complex vertical structure. To avoid false identification induced by multiple peaks between the ground and canopy peaks, a label-filling strategy is proposed. It fills the internal space between the ground and canopy peak locations with ones during the training process to learn both the peak profiles and the valid region of forest backscattering along the height direction. This label-filling operation forces our model to learn the ground and canopy peaks profile through the lower and upper envelopes of the tomogram, making it robust to these false peaks between the ground and canopy and able to accurately detect the ground and canopy peaks. Therefore, our model can be applied to the DTM and CHM estimation in tropical forest test sites.

After obtaining the 3-D backscattering profile and using the baseline configuration and radar systems parameters listed in Table I, one can simulate TomoSAR data through the following 2-D integration model [16]:

$$I_n = \sigma \int_{\Delta_z} s(z) \exp(-jK_n(z)) dz + \varepsilon_n \tag{3}$$

where I_n is the complex cross correlation between the nth flight and the reference flight; Δ_z represents the extent along the cross-range direction; $s(z)$ is the 3-D backscattering profile within this extent; $K_n(z)$ is vertical wavenumber between the reference track and the nth flight; and ε_n is a Gaussian random noise term. In addition, to make the simulation closer to real observation, we multiply the 3-D backscattering profile $s(z)$ with a power factor σ that corresponds to the amplitude of a real forest SAR image collected by a real SAR system. The introduction of σ here is to simulate the amplitude component, as the original 2-D integration model only simulates the phase component through observation geometry.

To demonstrate the effectiveness of simulation, a ground range line of the simulated scene is selected, and the corresponding 2-D Capon tomogram generated by the simulated TomoSAR dataset along this line is shown in Fig. 3, where the black and white dotted lines denote the ground and canopy peaks used for simulation, respectively. Except for some sidelobes, the lower and upper peaks of the tomogram generated by the simulated dataset fit quite well with the ground and canopy peak locations. Therefore, the simulated dataset can be used to train the 3-D U-net model to learn the ground and canopy peaks from a 3-D tomogram.

III. NUMERICAL RESULTS

In this section, the 3-D U-net model is trained on the simulated TomoSAR dataset. Then, the use of the label-filling strategy is justified through a real data test. After that, to validate the effectiveness of the trained model, the model is applied on multiple airborne P-band TomoSAR datasets to predict the DTM and CHM, and its performance is evaluated by comparing with the state-of-the-art Pol-TomoSAR techniques.

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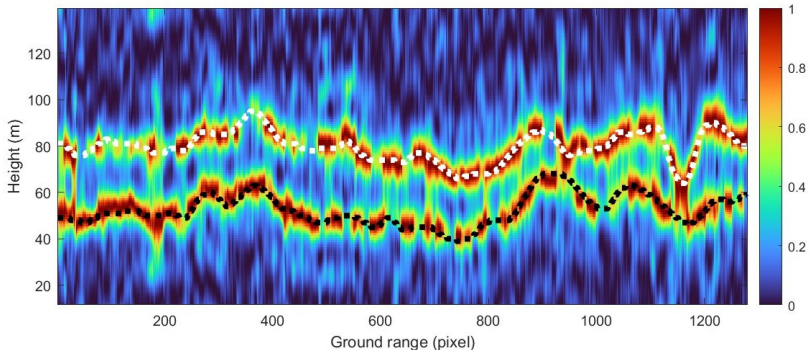


Fig. 3. 2-D Capon tomogram along a ground range line of the simulated single look complex (SLC) stack.

A. Model Training

To train the 3-D U-net model, the 3-D tomogram input data and the corresponding labels are tiled into small samples, and the size of each sample is $128 \times 128 \times 128$. Four hundred samples are generated in total, and they are randomly divided into a training dataset and a validation dataset in a ratio of 9:1. The Adam optimizer with adaptive learning rate is adopted to perform backpropagation parameter learning. The training samples are fed into the network in batches, and the batch size equals 2. In addition, to avoid possible model overfitting, a dropout layer is added at the end of each upsampling layer, which randomly shuts down 20% of the network links at each training step. This model is trained within the Google Colab environment with 53.0-GB RAM and NVIDIA L4 Tensor Core.

Fig. 4 shows the MAE loss curves with respect to total training steps. The black curve denotes the MAE loss with respect to training steps on the training dataset, and the red one is the MAE loss with respect to training epochs on the validation dataset. Each training epoch contains 180 training steps. It can be found that the model converges very fast within the first three epochs, and after that, the loss on both the training and validation datasets decreases at a much lower speed. In this article, we stop the training process once the loss on the validation dataset stops improving for three consecutive epochs.

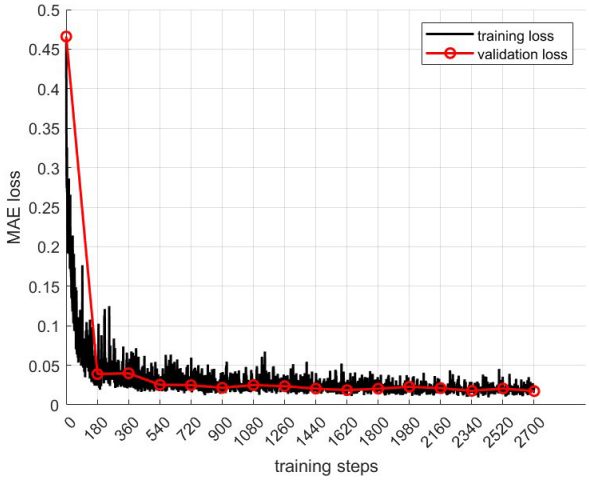


Fig. 4. MAE loss curves on training dataset and validation dataset.

B. Justification of the Label-Filling Strategy

It is mentioned in Section II-A that the label-filling strategy is vital for our 3-D U-net model. To justify this point, the model was trained with and without the label-filling strategy, respectively. Then the two models are tested on a small area of the Lope test site (see the data description in Section III-C). The HH polarization dataset is used, considering its better penetration into the forest. A transect along the range line, denoted as A in Fig. 7(a), is to be studied. Fig. 5(a) shows the Capon tomogram of transect A. Fig. 5(b) and (c) shows the predicted results of transect A by the model with and without using the label-filling strategy, respectively. The LiDAR DTM and CHM are also superposed on Fig. 5(b) and (c) with black and white solid lines, respectively. Fig. 5(d) and (e) shows the predicted ground and canopy peaks extracted from Fig. 5(b) and (c), respectively. The black and red dotted lines represent the extracted ground and canopy peak profiles, respectively. For comparison, the LiDAR DTM and CHM profile along transect A are also plotted in Fig. 5(d) and (e) using black and red solid lines, respectively. In the model using the label-filling strategy, the ground and canopy peaks are extracted by detecting the lower and upper envelopes of the predicted results. In the model without using the label-filling strategy, the ground and canopy peaks are extracted by detecting the two largest peaks of the predicted results.

It can be found in Fig. 5(c) that there are many false peaks between the actual ground and canopy peaks, which will lead to a strongly biased estimation if the two largest peaks are selected as the ground and canopy peaks. In Fig. 5(e), for example, the estimated ground profile by the model without the label-filling strategy shows very weak consistency with the LiDAR DTM profile. In Fig. 5(b), it can be observed that the lower and upper envelopes of the predicted results by the model using the label-filling strategy demonstrate much better consistency with the LiDAR DTM and CHM, except for a near constant offset between the LiDAR CHM and the upper envelope. The offset can also be found between the estimated canopy peaks profile and the LiDAR CHM in Fig. 5(d), which is a normal phenomenon as there exists a power loss between the canopy peak and the canopy top due to microwave

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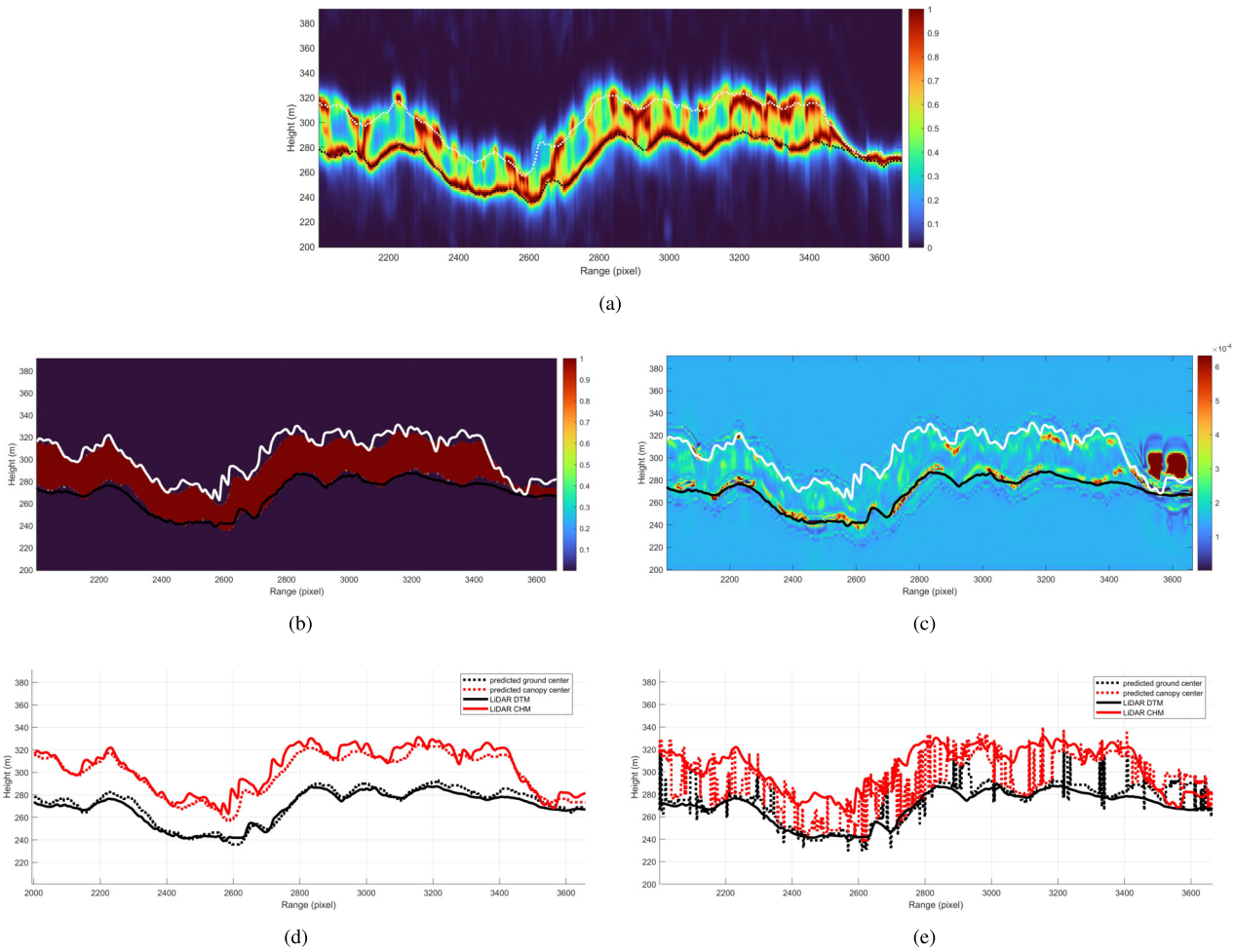


Fig. 5. Capon tomogram, predicated maps, predicted ground and canopy peak profiles of transect A. (a) Capon tomogram along transect A (the white and black lines represent the predicted canopy and ground peak profiles). (b) Predicted maps by the 3-D U-net model using label-filling strategy. (c) Predicted maps by the 3-D U-net model not using label-filling strategy. (d) Predicted ground and canopy peak profiles extracted from (a). (e) Predicted ground and canopy peak profiles extracted from (b).

attenuation [6], [20]. To demonstrate the performance of the model using the label-filling strategy in a more intuitive way, the predicted ground and canopy peak profile along transect A is superposed on the corresponding tomogram in Fig. 5(a) using a black and a white dotted line, respectively. It can be found that the identified ground and canopy profiles fit quite well with the lower and upper envelopes of the tomogram. Moreover, as shown in Fig. 5(d), the predicted ground and canopy peaks using the label-filling strategy show much better consistency with LiDAR references than the ones not using the label-filling strategy, which provides a powerful justification for the proposed label-filling strategy.

C. Real Data Experiments

1) Data Description: To test the effectiveness of the trained 3-D U-net model on a real TomoSAR dataset. Multiple P-band airborne TomoSAR datasets, collected in the AfriSAR campaign [21] by the German Aerospace Center (DLR) F-SAR system and the French Aerospace Lab (ONERA) SETHI system, are employed in the section. The radar system parameters of DLR's F-SAR and ONERA's SETHI are listed in Table II. And the corresponding LiDAR measurements acquired by NASA's land, vegetation, ice sensor (LVIS) [22] are used for performance evaluation in DTM and CHM estimation. The LVIS LiDAR footprint is over 20 m.

TABLE II
RADAR SYSTEM PARAMETERS OF F-SAR AND SETHI

Parameters	F-SAR	SETHI
Band	P-band	P-band
Carrier frequency	435 MHz	435 MHz
Bandwidth	50 MHz	50 MHz
Range resolution	3.9 m	3.0 m
Azimuth resolution	2.0 m	1.53 m
Polarization	Quad-polarization	Quad-polarization
Platform flight height	6096 m	6231 m
Incident angle	25°-55°	25°-55°

Three different test sites in Gabon, including Lope, Mabounie, and Rabi, are studied. The geographic locations of the three test sites are shown in Fig. 6.

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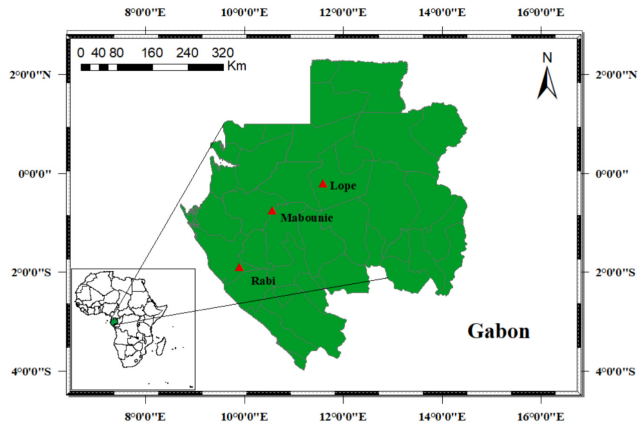


Fig. 6. Geographic locations of the three test sites.

TABLE III
BASELINE CONFIGURATION AT LOPE

Track	1	2	3	4	5	6	7	8	9	10
Baseline (m)	-80	-60	-40	-20	0	10	20	40	60	80

TABLE IV
BASELINE CONFIGURATION AT MABOUNIE

Track	1	2	3	4	5	6	7	8
Baseline (m)	-80	-60	-20	0	10	40	60	80

Lope test site (0.20°S, 11.59°E) is located in Lope National Park, and the topography ranges from 200 to 600 m. The area is basically covered by forest and savanna. The average tree height is over 40 m. F-SAR system collected ten full-pol images over this site during the AfriSAR campaign, and the baseline configuration is shown in Table III. The total baseline span is 180 m, and the average baseline spacing is about 20 m. Fig. 7(a) shows the PauliRGB SAR intensity image of the test site.

Mabounie test site (0.76°S, 10.56°E) is located in central western Gabon, and the topography ranges from 25 to 230 m. The study area is covered mostly by forest with over 30-m average tree height. DLR's F-SAR system took eight repeat-pass full-pol observations over this area during the 2016 AfriSAR campaign. The baseline configuration of the eight observations is listed in Table IV. The overall baseline span is about 160 m. Fig. 7(b) shows the PauliRGB SAR intensity image of the investigated area.

Rabi test site (1.92°S, 9.88°E) is located near the west coast of Gabon. The topography ranges from 30 to 80 m over the study area, and it is mostly covered by tropical forest with over 30-m average tree height. ONERA's SETHI system took ten repeat-pass full-pol observations over the study area during the 2015 AfriSAR campaign. The baseline configuration of the ten observations is shown in Table V. The overall baseline span is about 160 m, and the average baseline spacing is about

TABLE V
BASELINE CONFIGURATION AT RABI

Track	1	2	3	4	5	6	7	8	9	10
Baseline (m)	-80	-60	-40	-20	0	10	20	40	60	80

20 m. Fig. 7(c) shows the PauliRGB SAR intensity image of this site. Before this TomoSAR dataset is applied in DTM and CHM estimation, to remove the residual baseline phase errors, phase calibration [23], [24] is necessary and conducted using the calibration method in [23].

The LVIS LiDAR DTM and CHM data of the three test sites are back-geocoded into the corresponding radar coordinates for subsequent performance evaluation. Fig. 7(d)–(f) and (g)–(i) shows the back-geocoded LiDAR DTM and CHM maps of the three test sites, respectively. It can be noted that the spatial distribution of CHM maps looks consistent with the spatial distribution of radiometric distortion in SAR images because the LiDAR CHM maps in this article are mapped onto the radar range-azimuth coordinates from geographic longitude–latitude coordinates through a back-geocoding process, which means that if the tree is on a slope facing toward the radar, the actual tree height in geographic coordinates will be smaller than the tree height in radar coordinates. For the same tree, the radar tree height on a sloping ground facing toward the radar will also be larger than that on flat ground. The backscattering power of the target on a slope facing toward the radar is also stronger than that on flat ground, which is the reason why the CHM maps have similar patterns with radiometric distortion induced by terrain height variation in SAR intensity images. It can also be found that the LiDAR CHM in the radar coordinate can reach as high as 90 m, which actually corresponds to only about 45-m real tree height in geographic longitude–latitude coordinate for AfriSAR's radar incident angle.

2) DTM and CHM Estimation in Lope, Mabounie, and Rabi: The 3-D U-net model trained with a label-filling strategy is applied on the HH polarization TomoSAR datasets in three test sites to estimate DTM and CHM. Given that the Pol-TomoSAR method can achieve much better performance in DTM and CHM estimation than the single-pol TomoSAR method, the state-of-the-art Pol-TomoSAR techniques proposed in [5] and [9] are employed for DTM and CHM estimation performance comparison. Specifically, the Pol-TomoSAR method proposed in [9] is applied on Lope and Mabounie, and the one proposed in [5] is applied on Rabi.

For DTM estimation, the identified ground peak is exactly the estimated DTM. Fig. 8(a)–(c) shows the predicted DTM by the proposed model in the three test sites. For comparison, Fig. 8(d)–(f) shows the estimated DTM by the Pol-TomoSAR technique. In general, the DTM estimated by both methods demonstrates good visual consistency with the LiDAR DTM in Fig. 7. But if we take a closer look, it can be found that the Pol-TomoSAR DTM presents more biased estimations. For example, two regions of interest in Lope and Mabounie, as denoted by the red frames in Figs. 7(d) and (e), 8(a) and

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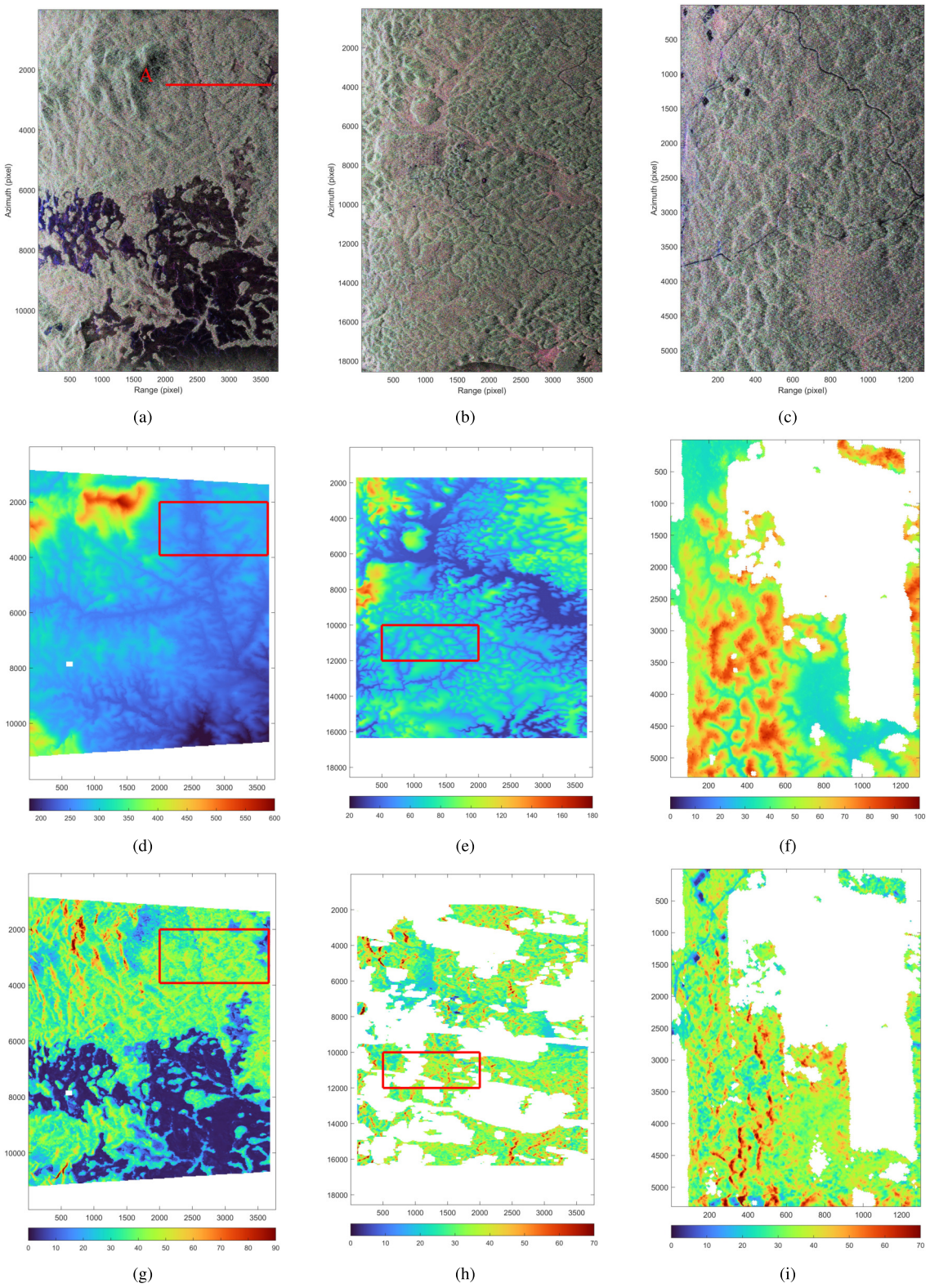


Fig. 7. PauliRGB SAR intensity images, LiDAR DTM and CHM of the three test sites. (a) PauliRGB SAR intensity images of Lope. (b) PauliRGB SAR intensity images of Mabounie. (c) PauliRGB SAR intensity images of Rabi. (d) LiDAR DTM of Lope. (e) LiDAR DTM of Mabounie. (f) LiDAR DTM of Rabi. (g) LiDAR CHM of Lope. (h) LiDAR CHM of Mabounie. (i) LiDAR CHM of Rabi.

(b), and (d) and (e), are selected and their enlarged views are shown in Fig. 9(a)–(f), respectively. Compared with the Pol-TomoSAR DTM, the predicted DTM presents much less bias and better consistency with the LiDAR DTM, which, according to [9], is the result of unsuccessful ground and volumetric components separation due to low GVR value. And

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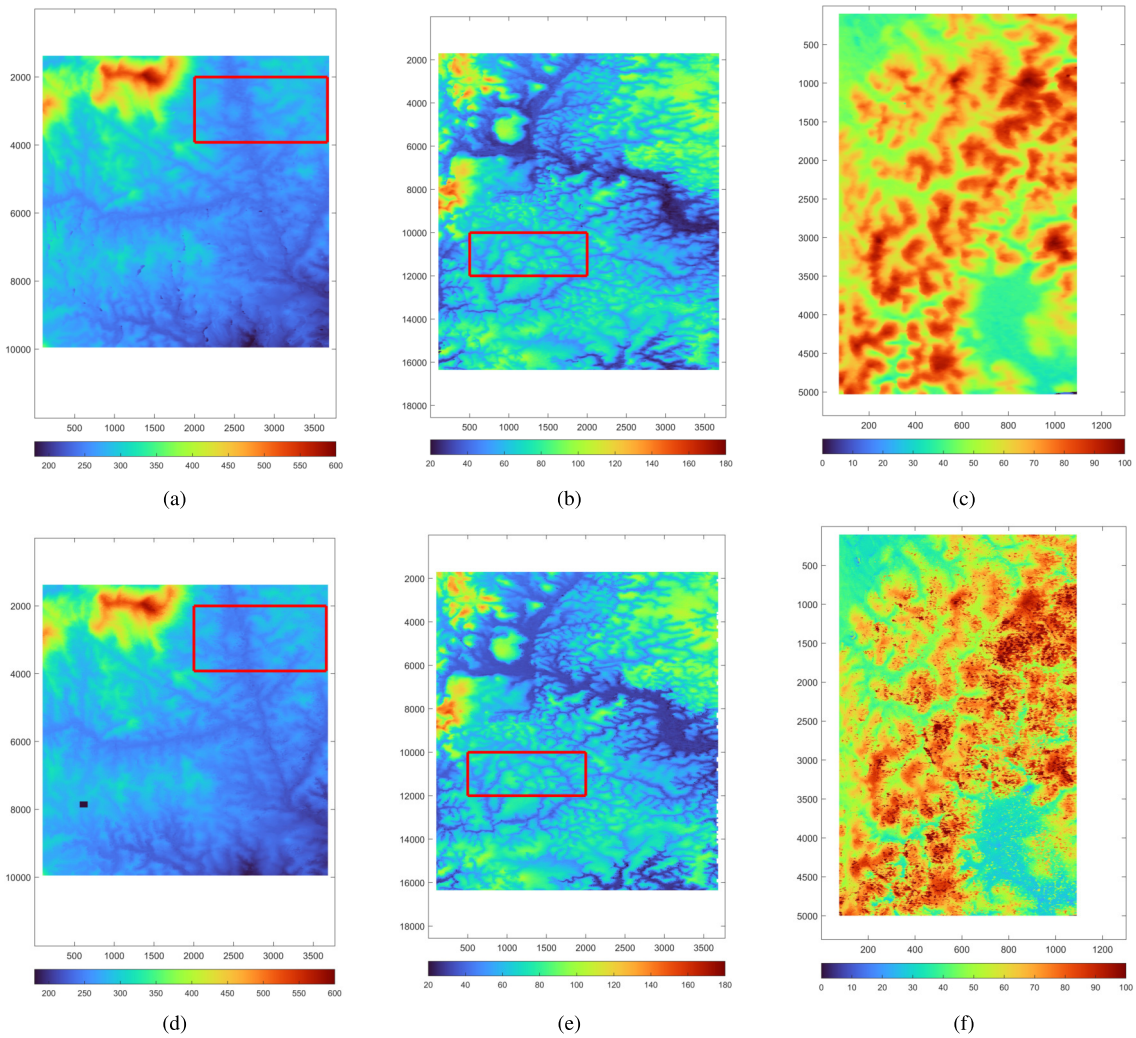


Fig. 8. DTM of the three test sites estimated by the proposed method and the Pol-TomoSAR method. (a) DTM of Lope predicted by the proposed method. (b) DTM of Mabounie predicted by the proposed method. (c) DTM of Rabi predicted by the proposed method. (d) DTM of Lope estimated by the Pol-TomoSAR method. (e) DTM of Mabounie estimated by the Pol-TomoSAR method. (f) DTM of Rabi estimated by the Pol-TomoSAR method.

these biased estimations are no longer visible in the predicted DTM because the proposed deep-learning model can learn the patterns of ground locations from 3-D tomograms even with a rather low GVR. To perform a quantitative evaluation, the joint distributions between the estimated DTM by the two methods and the LiDAR DTM of the three test sites are shown in Fig. 10(a)–(c) and (d)–(f), respectively. Note that the distributions are normalized such that the maximum equals one. And accordingly, the root-mean-square errors (RMSE) and coefficient of determination (R^2 value) with respect to the LiDAR measurements are also computed on a pixel basis. Compared with the Pol-TomoSAR technique, obvious improvements can be found in the predicted DTM by the proposed model at all three test sites. For Lope, the RMSE improves from 5.7 to 3.4 m, with 40.4% improvement; the R^2 value improves from 0.9906 to 0.9968, with 0.6% improvement. For Mabounie, the RMSE improves from 4.9 to 3.9 m, with 20.4% improvement; and the R^2 improves from 0.9337 to 0.9579, with 2.6% improvement. For Rabi, the RMSE improves from 11.0 to 3.7 m, with 66.4% improvement; and the R^2 value improves from 0.6224 to 0.9361, with 50.4% improvement. From Fig. 10(f), we can notice a very pronounced overestimation of Pol-TomoSAR DTM in Rabi, which is a sign of unsuccessful ground separation by the Pol-TomoSAR method in [5]. According to [9], these unsuccessful separations often happen in areas with large negative slope angles, where the ground response power is much lower than in other areas. In these areas, the separated ground components are often dominated by residual canopy volumetric components, which will result in an overestimation if they are used for DTM estimation.

From Fig. 7(f), Fig. 8(f) and (c), we can notice a slight overestimation on the upper left corner of the DTM by the proposed deep-learning method because the tree height is lower than the TomoSAR resolution limit. This overestimation, however, is not presented in the DTM by the Pol-TomoSAR method because the Pol-TomoSAR method uses full-pol data, and it can partially overcome the inherent resolution limit as it can separate the ground and volumetric scattering components by exploring the difference between their polarimetric characteristics even if the spatial distance between the two scattering centers is smaller than the resolution limit.

In terms of the CHM estimation, as mentioned in many literatures [6], [9], [20], there exists a power loss offset

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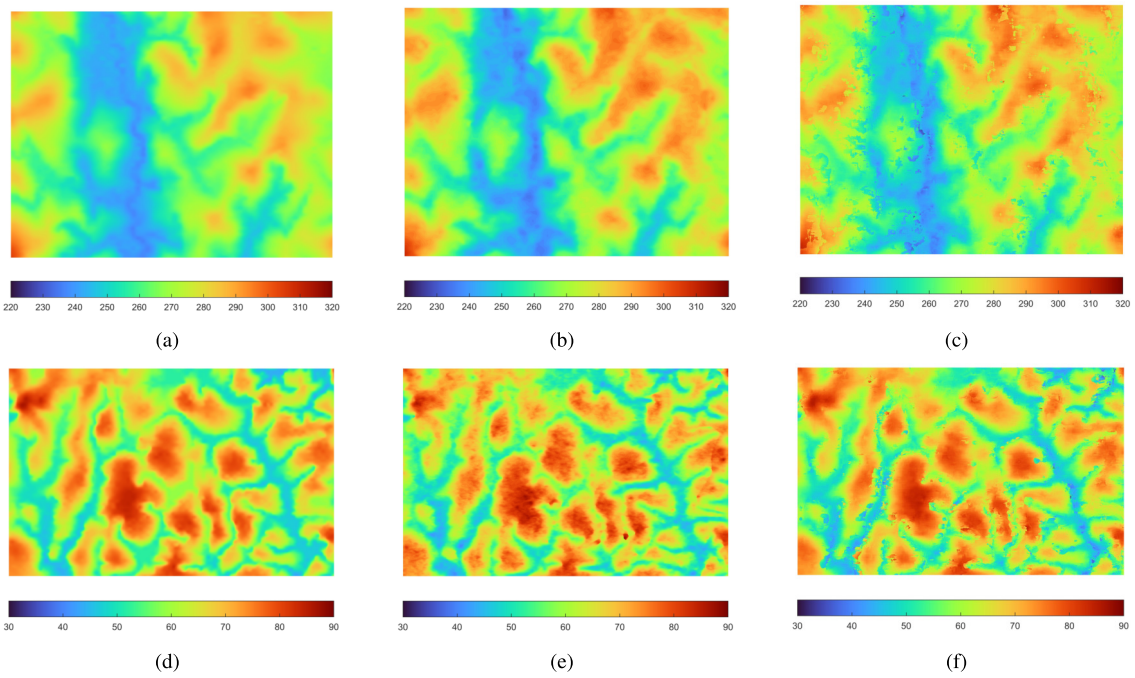


Fig. 9. Enlarged views of DTM of Lope and Mabounie estimated by the proposed method and the Pol-TomoSAR method. (a) Enlarged view of the red frame in Fig. 7(d). (b) Enlarged view of the red frame in Fig. 8(a). (c) Enlarged view of the red frame in Fig. 8(d). (d) Enlarged view of the red frame in Fig. 7(e). (e) Enlarged view of the red frame in Fig. 8(b). (f) Enlarged view of the red frame in Fig. 8(e).

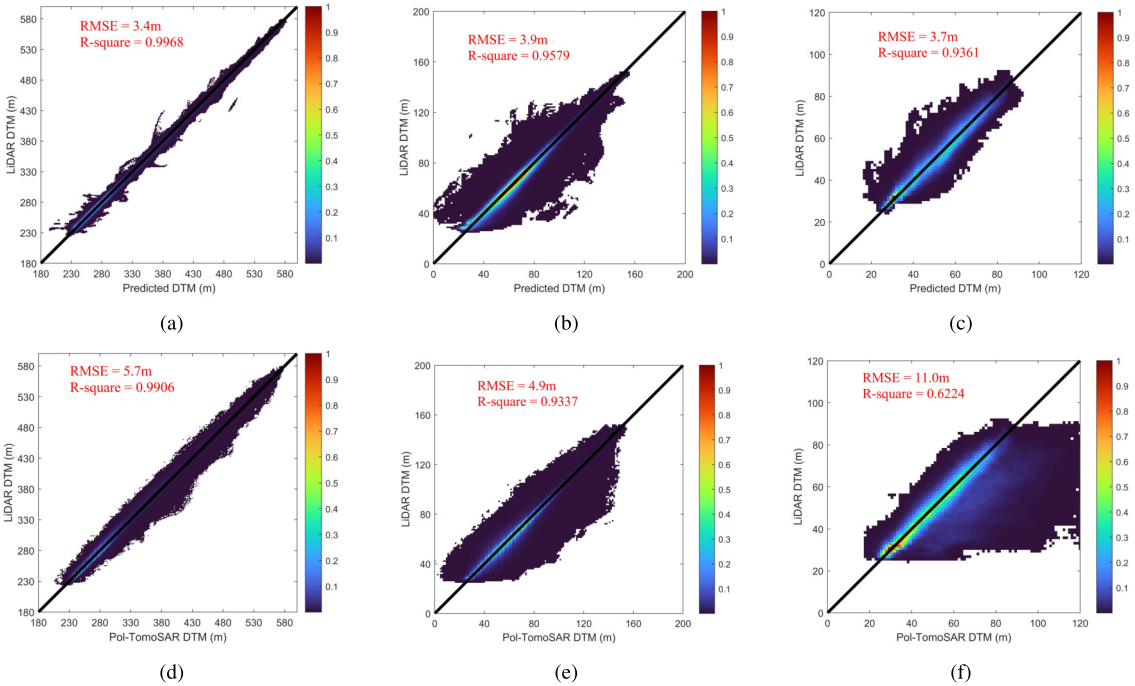


Fig. 10. Joint distributions between the LiDAR DTM and the DTM estimated by the two methods in the three test sites. (a) Proposed method at Lope. (b) Proposed method at Mabounie. (c) Proposed method at Rabi. (d) Pol-TomoSAR method at Lope. (e) Pol-TomoSAR method at Mabounie. (f) Pol-TomoSAR method at Rabi.

between the canopy top (CHM) and the canopy phase center due to microwave attenuation. Therefore, the detected canopy peaks cannot be treated as CHM directly, and usually, the LiDAR measurement is employed to calibrate this offset. In this article, the offset between the CHM and the detected canopy peak is treated as a constant and calibrated by the LiDAR CHM. Fig. 11(a)–(c) shows the predicted CHM by the proposed model of the three test sites. Fig. 11(d)–(f) shows the Pol-TomoSAR CHM of the three test sites. Compared with the LiDAR CHM in Fig. 7, the estimated CHM by both methods demonstrates not bad consistency with LiDAR measurement. To get a closer view, two regions of interest in Lope and

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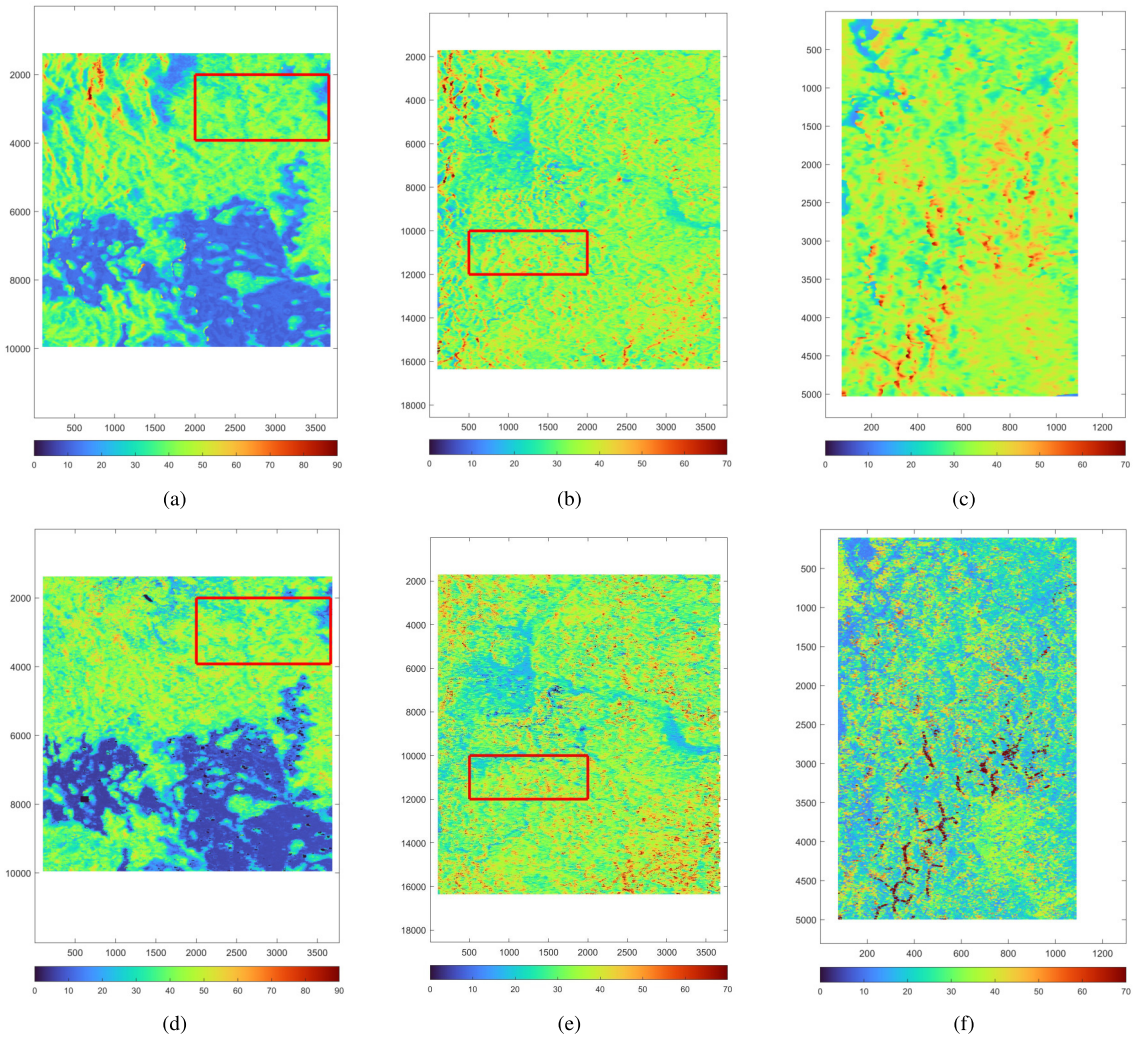


Fig. 11. CHM of the three test sites estimated by the proposed method and the Pol-TomoSAR method. (a) CHM of Lope predicted by the proposed method. (b) CHM of Mabounie predicted by the proposed method. (c) CHM of Rabi predicted by the proposed method. (d) CHM of Lope estimated by the Pol-TomoSAR method. (e) CHM of Mabounie estimated by the Pol-TomoSAR method. (f) CHM of Rabi estimated by the Pol-TomoSAR method.

Mabounie, as denoted by the red frames in Figs. 7(g) and (h), 11(a) and (b), and 11(d) and (e), are selected and their enlarged views are shown in Figs. 12(a)–(f), respectively. It can be found immediately that the Pol-TomoSAR CHM presents much more biased estimations than the predicted CHM by the proposed model. And generally, the predicted CHM demonstrates better consistency with the LiDAR CHM. To perform a quantitative evaluation, the joint distributions between the estimated CHM by the two methods and the LiDAR CHM of the three test sites are shown in Fig. 13(a)–(c) and (d)–(f), respectively. And accordingly, the RMSE and coefficient of determination with respect to the LiDAR measurements are also computed on a pixel basis. Compared with the Pol-TomoSAR technique, obvious improvements can be found in predicted CHM by the proposed model at all three test sites. For Lope, the RMSE improves from 8.3 to 4.9 m, with 41.0% improvement; the R^2 value improves from 0.5744 to 0.8352, with 45.4% improvement. For Mabounie, the RMSE improves from 7.8 to 5.1 m, with 34.6% improvement; and the R^2 improves from 0.3226 to 0.6084, with 88.6% improvement. For Rabi, the RMSE improves from 15.2 to 5.6 m, with 63.2% improvement; and the R^2 value improves from 0.1268 to 0.5380, with more than three times improvement. From Fig. 13(f), we can notice a very pronounced underestimation of Pol-TomoSAR CHM in Rabi, which is also a sign of unsuccessful ground separation by the Pol-TomoSAR method in [5]. According to [9], these unsuccessful separations will result in an overestimation if they are used for DTM estimation. Subsequently, we will get an underestimated CHM because CHM is obtained by subtracting DTM from the canopy top height. And this problem was largely mitigated in [9]. Therefore, this underestimation is not presented in Fig. 13(d) and (e).

One important aspect to evaluate a deep-learning method is its generalization ability. Our model is trained on a simulated dataset of seven images with 15-m baseline spacing and tested on real TomoSAR data from three different test sites of the AfriSAR campaign, which has at least eight images with 20-m baseline spacing for each test site. Despite such different baseline configurations, results demon-

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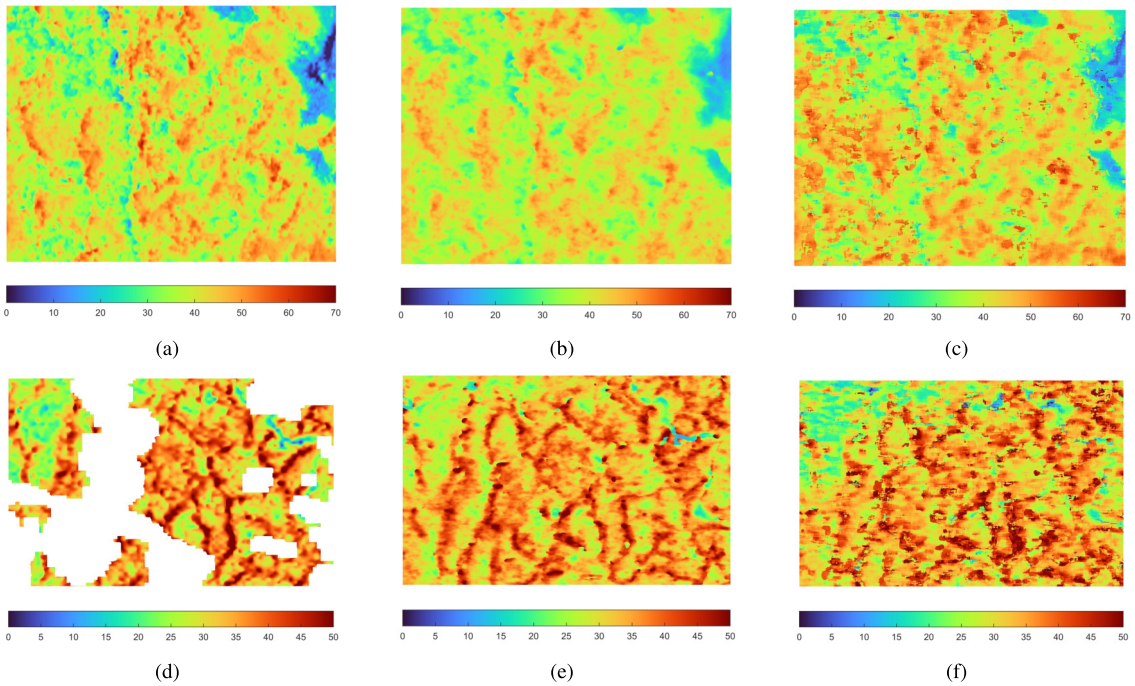


Fig. 12. Enlarged views of CHM of Lope and Mabounie estimated by the proposed method and the Pol-TomoSAR method. (a) Enlarged view of the red frame in Fig. 7(g). (b) Enlarged view of the red frame in Fig. 11(a). (c) Enlarged view of the red frame in Fig. 11(d). (d) Enlarged view of the red frame in Fig. 7(h). (e) Enlarged view of the red frame in Fig. 11(b). (f) Enlarged view of the red frame in Fig. 11(e).

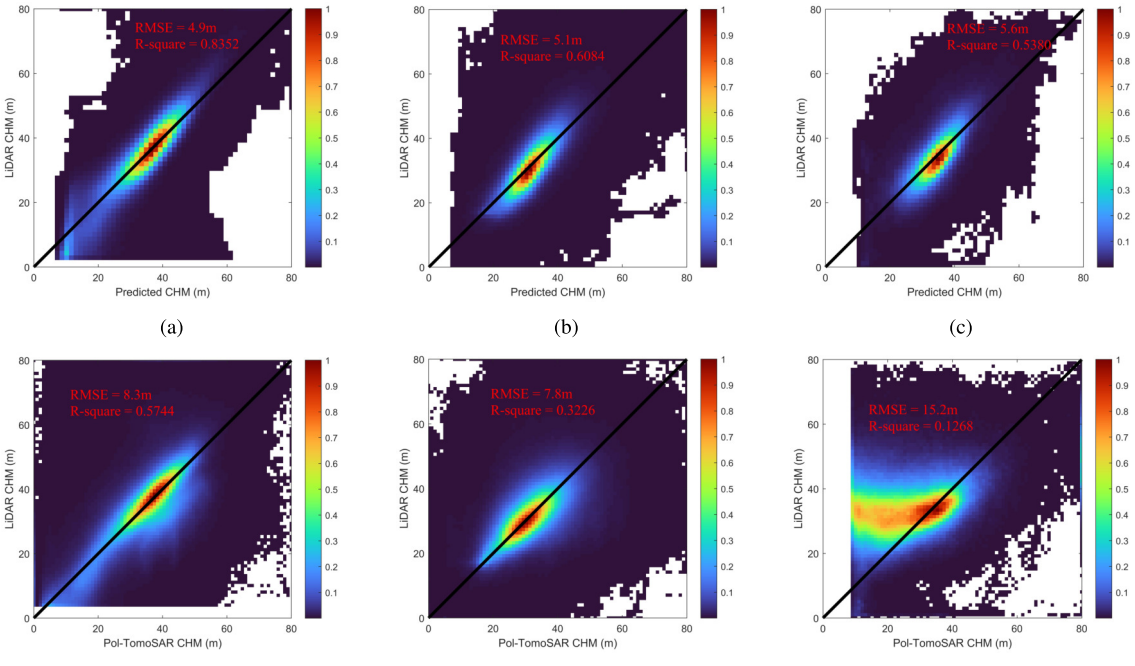


Fig. 13. Joint distributions between the LiDAR CHM and the CHM estimated by the two methods in the three test sites. (a) Proposed method at Lope. (b) Proposed method at Mabounie. (c) Proposed method at Rabi. (d) Pol-TomoSAR method at Lope. (e) Pol-TomoSAR method at Mabounie. (f) Pol-TomoSAR method at Rabi.

strate that our model can well adapt to these different baseline distributions from the training scenario and subsequently give more accurate DTM and CHM estimation than the traditional Pol-TomoSAR method. This shows that our method is able to generalize in scenarios of different baseline distributions.

To further compare the computational burden of the proposed deep-learning method and the Pol-TomoSAR method [9], a 128×128 window is selected for DTM and CHM inversion, and the corresponding running times of the two methods are listed in Table VI. In this case, both methods were running in CPU mode on a computer with 32-GB RAM

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TABLE VI
PROCESSING TIME OF THE TWO METHODS ON A 128 × 128 WINDOW

Method	Proposed	Pol-TomoSAR
Processing time (seconds)	1.5	11.3

and an Intel Core i9-13900H 2.60-GHz CPU. It can be found that the processing time of the proposed deep-learning method is much less than the Pol-TomoSAR method, demonstrating its better computational efficiency.

IV. CONCLUSION AND DISCUSSION

In this article, a novel method based on a 3-D U-net deep-learning model is proposed to identify the ground and canopy peaks from single-pol TomoSAR 3-D tomograms. There are two key features of the proposed method. The first one is that the model can be well trained using only a simulated TomoSAR dataset, which makes it easy to implement when we do not have enough real TomoSAR datasets with LiDAR references. The other one is the label-filling strategy used on model training. Rather than just labeling the ground and canopy peak locations in the tomograms as one and the others as zero, we propose filling the internal locations between the ground and canopy peaks with ones as well. This operation can take full advantage of the capability of the 3-D U-net structure in both local and global feature extraction and learning. Real experiment conducted on Lope TomoSAR dataset has demonstrated that, compared with the one not using label-filling strategy, the model using label-filling strategy can greatly improve the performance in ground and canopy peaks identification. Then the model was applied to three AfriSAR campaign test sites, including Lope, Mabounie, and Rabi, for DTM and CHM estimation. The results show that, compared with the state-of-the-art Pol-TomoSAR techniques, the proposed method can make a large improvement in DTM and CHM estimation accuracy using only single-pol observation.

The potential improvement that can be made in the future about the proposed method is in the TomoSAR vertical resolution. Efforts can be made to overcome the resolution limit, such that we can learn the ground and canopy peak locations even from tomograms generated by insufficient TomoSAR observations.

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